Device Manager

Endpoint / cluster discovery

Configurations and capabilities

Message / attribute support

Device Profile Store
Target scope · endpoints · clusters · capabilities

4. Protocol State Awareness

4.1 Multi-Layer Message Analysis

Endpoint Analysis → Format Description Extraction → LLM-Based Format Generation

4.2 Message Relationship Analysis

Message Dependency Collection + Message Correlation Discovery

4.3 Protocol State Exploration

State Graph Construction

Potential State Transition Discovery

Protocol / Fuzzing Graph

5. Device State Awareness

5.1 State Attribute Collection

Stage 1
Active Query

Stage 2
Static Analysis
Specs + Docs

Stage 3
Hidden State Discovery

Complex Data-Type Parsing
Basic and hidden state attributes

5.2 Attribute Permission Analysis

State Attributes
Read / write method
Message link
Setup candidates

6. State-Guided Fuzzer

6.1 Mutator Agent · ReAct Test Case Generator

State-Guided Protocol Messages Generation Tool

Correlated Device State Analysis Tool

Format-Aware Mutation Tool

Two-Level UCB Tool

Prompt Context Builder
Profile + runtime state
Retrieved corpus

SOTA LLM Mutator Brain

Result Check
Format / state validation

Real Testbed

Gateway Sender

1. SET Device State
+
2. SEND Message Sequence

IoT Device

6.2 Monitor Agent

Traffic and response
Liveness and snapshots
Crash / anomaly evidence

Reads / writes Runtime State Store

6.3 Evaluator Module

Impact · progress · cost
Replay-confirmed anomaly
Sequence-level credit

Updates UCB + long-term corpus

Persistent State and Agent Memory

Short-Term Memory
Recent actions and response trace
Rolling context and outcomes

Long-Term Memory / Corpus
High-value payloads and replayable sequences
Evaluator writes · Context Builder retrieves

Runtime State Store
Current device state and liveness
Monitor reads / writes · Mutator reads

UCB Statistics Store
Message / strategy counts and returns
Evaluator writes · UCB Tool reads